Retrieval Relevance Is Not Verification Priority: Budgeted Provenance Checking in Inherited Agent Memory

Anonymous Author(s)
Affiliation
Address
email

Abstract

A memory system that gives every record a provenance link is not thereby safe. An agent inheriting hundreds of consolidated beliefs can follow only a few links before it acts, so the safety-critical question is not whether provenance exists but which links get followed. We study that allocation directly. In a controlled environment where exactly one of six inherited memories has lost a true negative caveat, and the agent may pull at most k source records before committing, we find allocation is governed by the agent's own plan: across 1020 pre-registered episodes the corrupted memory is verified in 236/236 episodes whose first-pass intent it backs and 464/784 otherwise — complete separation, no counterexample. Two new pre-registered experiments then establish that this allocation causes later behaviour rather than merely correlating with it. Randomising which source records are carried into a later decision, after the budget is spent and the situation has shifted, moves the corrupted-direction rate from 139/150 to 3/150 — 91 points pooled, and the same direction in all 6 models. A third experiment separates what drives allocation from what the memory means: with the agent's plan held fixed, a memory carrying an *irrelevant* hedge is checked more often than the same memory with its caveat silently deleted, so omission is stealthier than qualification. Finally, we test the premise the threat model rests on by running a real consolidation chain, and it fails: over six generations of re-summarisation, quantified negative outcomes mostly survive (83% of chains at the tightest compression, 6/10 to 10/10 per model), while scope restrictions and normative prohibitions erode. The corruption that matters is loss of the constraint, not loss of the fact. We argue persistent-memory systems need a verification scheduler distinct from the retriever, and that the two rank records differently.

1 Introduction

Long-horizon agents do not reason from evidence. They reason from memory: lessons written by earlier sessions, compressed by consolidation passes, and inherited as durable organisational knowledge. Provenance-aware memory designs answer this by keeping a link from every consolidated record back to its source, so a suspicious belief can be checked.

That defence has a budget. An agent that inherits hundreds of beliefs and must act this cycle can follow two or three links, not two hundred. Whether the memory system is safe therefore depends on a decision the retrieval layer never makes: given many records that could have drifted and no labels saying which did, *which records does the agent choose to audit?*

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

This is a ranking problem, and it is not the ranking a retriever computes. A retriever ranks by relevance to the current query or plan. What an audit should rank by is the expected cost of the record being wrong — which is often highest exactly where the agent is *not* currently looking, because a belief the agent is about to act on will be tested by acting, while a belief sitting quietly in memory will not.

We study this gap in a controlled setting. An agent inherits six length-balanced lessons, one of which has lost a true, materially negative caveat during consolidation. It may retrieve at most k source records before committing to an action. Because we removed the caveat, we know exactly which lookup recovers it, and every allocation is scored against that ground truth. The corrupted belief is deliberately *not* the one the situation makes salient: the symptoms point toward onboarding work, while the corrupted memory recommends promotional pricing.

Contributions.

1. **Problem formulation.** Budgeted selective verification over inherited memory as a reliability variable distinct from retrieval and from detection (§3).
2. **Measurement.** Verification allocation is governed by the agent's current intent, with complete separation over 1020 pre-registered episodes (§4).
3. **Causal identification.** Two pre-registered experiments — a randomised-instrument replay and a randomised carry-forward — show that what the budget recovered *causes* the later decision, with the confounds of the observational design removed by construction (§5).
4. **Mechanism.** With intent held fixed, allocation responds to unresolved qualification, and a silently deleted caveat attracts *less* scrutiny than an irrelevant hedge (§6).
5. **A negative result that reframes the threat model.** Real consolidation does not produce the corruption this literature assumes. Over six generations, quantified negatives survive; prohibitions erode (§7).

Study 1 (§4) re-analyses episode files released with an earlier preprint by the same authors; the citation is omitted here for anonymity and will be restored in the camera-ready. Studies 2–4 are new, pre-registered before any model call, and reported with their failures.

2 Related work

Memory integrity. Memory-poisoning work studies adversarial insertion into agent memory [Dash et al., 2026, Xie et al., 2026] and defences that trust sources rather than content [Singh, 2026]. Our corruption is not adversarial: nothing false is inserted, and the drifted summary is what a benign compression pass produces. The threat model is operational, extending fault injection [Tan et al., 2026] from transport faults to the semantic layer.

Budgeted verification. A growing line allocates limited verification compute across candidate *outputs* or reasoning steps [Yuan et al., 2026a, Fang et al., 2026, Yang, 2026], and benchmarks ask whether agents manage tool budgets at all [Wang and Xu, 2026, Lin et al., 2026]. We allocate over *inherited beliefs* and measure the direction of the allocation rather than its total.

Provenance and self-auditing. Execution-provenance surveys catalogue how evidence and actions connect in agent traces [Wang et al., 2026]; self-auditing repairs unsupported intermediate beliefs [Yuan et al., 2026b], and provenance-sensitivity audits ask whether action selection responds to source information at all [Liao, 2026]. These assume the audit target is given, or audit everything. We study the upstream selection problem.

Positive test strategies. That an agent checks the evidence behind its favoured hypothesis rather than evidence that could overturn it rhymes with human hypothesis-testing research [Wason, 1960, Klayman and Ha, 1987]. We claim no shared mechanism; we claim the allocation signature is measurable and steerable in agents.

Table 1: Firth-penalised logistic model of whether the corrupted memory's source was requested, over all 1020 Study-1 episodes. Robustness and synthesis; the pre-registered contrasts remain the confirmatory analysis.

term	odds ratio	95% CI
intent aligned	422	[26, 6860]
caveat deleted	4.44	[2.73, 7.20]
triage invariant	13.2	[7.60, 23]
budget (+1 credit)	13.2	[6.98, 24.8]
target position	0.99	[0.89, 1.11]
corruption moved onto the intended path	3.90	[1.13, 13.5]

3 Setup

A growth agent for a synthetic B2B SaaS business inherits six durable memories, each a one-to-two-sentence lesson with a consolidation date and an archived source. Bodies are length-balanced (123–132 characters) and identically formatted; presentation order is shuffled per episode with a seeded PRNG. The situation presents three concurrent symptoms so that four of the six memories bear on something and no action trivially dominates. The agent may retrieve at most k source records by id, then commits to one action. Six models from two providers are used throughout, at their default or medium reasoning setting, with byte-identical prompts and structured output enforced.

In the *clean* arm the target memory carries its true caveat; in the *drifted* arm the caveat is absent and the remaining sentence is still true. A *triage* arm adds one system-prompt sentence: *when verification capacity is limited, prioritize inherited beliefs that are both highly consequential to the current decision and insufficiently supported for the objective being optimized*. It names no memory, metric, or direction.

4 Study 1: allocation follows intent

Across all 1020 episodes — every condition, budget, target and model — the corrupted memory is verified in **236/236** episodes whose first-pass intended action it backs, against 464/784 (59%) otherwise. The relation is completely separated: the corpus contains no episode in which the agent planned to act on the corrupted belief and did not check it.

Table 1 fits one model to all 1020 episodes. Because intent-alignment separates the outcome perfectly the ordinary MLE diverges, so we use Firth's penalised likelihood; the separation is the finding, not a nuisance. Target position carries an odds ratio of 0.99, confirming that randomising presentation order removed it as a confound.

A uniform policy at $k{=}2$ over six memories reaches the target 33% of the time. Models are better than that on average and far better when the target is in path — and worse than uniform in the models at the bottom of the spread. The aggregate hides the structure: allocation is not weakly risk-sensitive, it is strongly plan-sensitive.

5 Study 2: the allocation causes the later decision

Study 1 shows where the budget goes. It does not show that this matters. The natural test — let the situation shift after the budget is spent, then see whether episodes that recovered the caveat behave differently — is observational: verification is chosen by the model, so the two groups differ in whatever made the model choose.

An earlier two-phase experiment ran exactly that comparison and reported a large split. Auditing its implementation we found two paths that the design left open and the reporting did not separate. First, the triage arm's system prompt was constructed once and reused for the later call, so the randomised instruction was still in context when the second decision was made and could act on it directly. Second, the phase-1 conversation was carried forward, so the model saw its own earlier reasoning — which in the drifted arm frequently argued explicitly against the corrupted direction.

Table 2: Study 2b. Corrupted-direction rate when the provenance carried into the later decision is randomised rather than chosen. Both arms carry two source records; only whether one of them is the corrupted memory's differs.

	corrupted memory's source carried	not carried
Opus 5	0/25	14/25
Sonnet 5	2/25	25/25
Haiku 4.5	0/25	25/25
Sol	0/25	25/25
Terra	1/25	25/25
Luna	0/25	25/25
pooled	3/150 (2%)	139/150 (93%)

"Verified" and "unverified" episodes therefore differed in what the model had already committed to in writing, not only in what evidence it held.

We close both by construction, pre-registering two experiments before any call.

5.1 A randomised-instrument replay

For each of the 450 episodes of the earlier experiment we replay *only* the second decision, as a fresh session: the triage sentence is absent from every arm, no conversation history is carried, the memory block and its per-episode order are reproduced, and the only thing that differs is a RECOVERED block holding source records for exactly the memories that episode's budget was actually spent on. The situation text is byte-identical across arms.

Triage assignment was randomised upstream and moves recovery hard. In the replay it cannot reach the outcome by any route except through what was carried forward, because the session is otherwise identical across arms. This is a randomised encouragement design in which the exclusion restriction is true by construction rather than assumed.

It replicates. The corrupted-direction rate is 13/150 (9%) under triage against 55/150 (37%) without it (CMH stratified by model, $p = 2.2 \times 10^{-9}$). And the exclusion restriction can be checked rather than argued: matched on what phase 1 recovered, the two arms are indistinguishable in both strata — 4/94 versus 3/140 among episodes that recovered the caveat ($p = 0.764$), and 51/56 versus 10/10 among those that did not ($p = 0.61$). With the instruction removed from the room, arms holding the same evidence behave the same way.

The Wald estimate of the effect of recovering the corrupted memory's source is -0.91 (95% CI $[-1.34, -0.62]$ by stratified bootstrap over models): close to complete compliance, an agent that recovers the caveat essentially does not walk into the corrupted direction.

The original design was conservative, not inflated. Removing the carried-forward conversation *raises* the corrupted-direction rate among episodes that never recovered the caveat, from 79% in the original to 91% here. The model's own earlier prose had been suppressing the choice, and the original design credited that suppression to provenance. The confound worked against the reported effect.

5.2 Randomising the provenance itself

The replay still lets the model choose what it recovered. So we also assign it. In 300 fresh single-call episodes the memory block is the drifted lineage in both arms, and two source records are carried forward: in CARRY-TARGET one of them is the corrupted memory's, in CARRY-OTHER neither is. Both arms carry exactly two records; only their identity differs, and it is assigned by seeded PRNG.

Table 2 gives the result: 3/150 against 139/150, a difference of 91 percentage points (CMH $\chi^2 = 246$, $p < 10^{-54}$), in the same direction in 6 of six models, with five of six choosing the corrupted direction in every single episode where its provenance was absent.

Table 3: Study 3. Verification of the corrupted memory, first pass, budget two, restricted to episodes where the agent does *not* intend the action that memory backs. Bodies are matched on length and clause structure across the three appended arms; only the content differs.

	clean-negative (true caveat)	drifted (no qualifier)	clean-neutral (irrelevant hedge)	clean-positive (positive)
verified	37/150 (25%)	60/123 (49%)	82/129 (64%)	51/71 (72%)

Two independent identification strategies — an instrument in 2a, full randomisation in 2b — put the effect at the same place, near -91 percentage points. What the budget happened to recover is not a marker of some other difference between episodes. It is the thing that decides.

What this does and does not establish. It establishes that *holding* the recovered caveat changes the later decision. It does not establish that models allocate budget *in order to* secure future evidence; Study 1 says the opposite, that they allocate toward the present plan. The causal chain we have closed is allocation $\rightarrow$ what is in context later $\rightarrow$ behaviour, not any claim about foresight.

6 Study 3: what allocation actually responds to

Study 1 contrasts a drifted memory against a clean one carrying its caveat and reads the difference as detection of the omission. Auditing the construction we found that contrast varies four things at once. Appending the caveat makes the target body 221 characters against 123–132 for every other memory — $1.73\times$ — and makes it the only memory in the set carrying a *However* clause. Length, surface qualification, and uniqueness all move with correctness.

So we separate them. Four arms run concurrently, differing only in what is appended to the corrupted memory's body, matched to the true caveat on character length and clause structure: **drifted** (nothing appended, 129 characters), **clean-positive** (a positive elaboration, no hedge, 220), **clean-neutral** (a measurement-hygiene note that weakens confidence in the finding without saying the action is risky, 221), and **clean-negative** (the true caveat, 221).

The arms are not intent-matched, and that has to be handled. Our controls changed what the agent plans to do. Appending a positive elaboration made pricing more attractive — 79/150 episodes intend it, against 27/150 under drift — and the true caveat removed the plan entirely (0/150). Since intent alignment is the dominant driver of verification (§4), the marginal comparison across arms is confounded by it. We therefore report the intent-misaligned stratum, where the agent does not plan to price and the qualifier is the only thing that varies. This costs power in the **clean-positive** arm, which loses half its episodes.

Table 3 orders the arms in a way a salience account does not predict, and the ordering is monotone in something other than how alarming the text is. The true caveat does not raise verification; it *collapses* it to 25%. A matched-length qualifier that raises a question without answering it raises verification (64% and 72%; 9.7×10^{-14} and 1.0×10^{-13} against the true caveat). The silently drifted memory sits between them at 49%.

The mechanism is visible in where the budget goes instead. In the arm carrying the true caveat, one model's first-pass intent was onboarding in every episode and its two credits went to the memories backing that plan; the corrupted memory was never requested in 25 episodes. Nothing about that memory had become less important — what changed is that its constraint was already written down, so there was nothing to recover.

Two things follow.

The published contrast holds, but not for the stated reason. Verification really is higher when the caveat is absent than when it is present, and it survives intent-matching (60/123 against 37/150, 2.0×10^{-6}). But this is not the agent noticing that something was removed. It is the agent not needing to look at a belief whose constraint is already stated. Detection of the omission is not required to produce the effect, and Study 3 provides no evidence that it occurs. Study 1's reading of

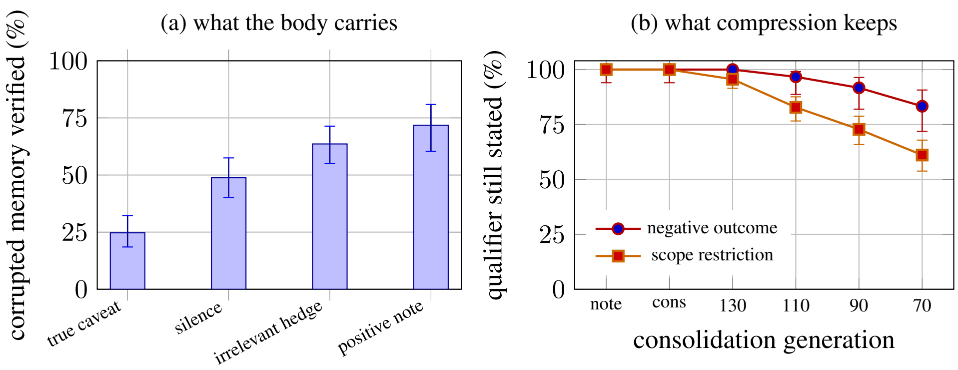

Figure 1: **(a)** Verification of the corrupted memory by what its body carries, restricted to episodes where the agent does not intend the action that memory backs (Study 3). The true caveat suppresses verification; a qualifier that raises a question without answering it raises it; silence sits between. **(b)** Survival of each qualifier type through six consolidation generations (Study 4); x-axis is session note, consolidation, then re-summarisation targets of 130, 110, 90 and 70 characters. Scope restrictions erode; quantified negative outcomes largely do not. Bars are Wilson 95% intervals.

this contrast as detection — and its interpretation of one model's reversal as a "verify-me flag" — is not supported once length and qualification are controlled.

Silence is stealthier than qualification. A memory that hedges about something irrelevant draws more scrutiny than one whose material caveat was deleted (15 points, 0.004). Consolidation that drops a clause therefore does not merely remove information — it removes the surface cue that would have made the record look worth checking. The record becomes cleaner-looking and correspondingly less audited, which is the worst combination for a store that future sessions inherit. Study 4 shows this is exactly the direction real consolidation moves in.

7 Study 4: does benign consolidation actually produce this?

Studies 1–3 install the corruption by hand, which buys clean ground truth. It also assumes something the literature repeats and, as far as we can tell, has not measured: that summarisation drops caveats. We tested it, expecting confirmation, and did not get it.

We ran a real consolidation chain over the same six source records, with no instruction to preserve or drop anything and no mention of caveats, qualifiers, or what matters: an analyst writes session notes; the notes are consolidated into durable lessons; the durable memory is then re-summarised four times as it ages and later cycles compete for context, with targets stepping 130 $\rightarrow$ 110 $\rightarrow$ 90 $\rightarrow$ 70 characters. 60 chains, six generations each, across six models.

Survival is scored by a model that sees the original source record and all six generations at once and reports, for each, whether the source's specific constraint is still stated. Two of the six records carry no qualifier at all and act as a false-positive check: the scorer identified them as unqualified in every case and reported a constraint in $0/120$ judgments where none exists.

The premise fails for negative outcomes. A quantified negative result is close to the last thing compression discards, though not immune. At the tightest compression — lessons cut to around 70 characters — the retention harm is still stated in 83% of chains, ranging from 6/10 to 10/10 across models. Several models reach that budget with sentences of the form "40% SMB discount: signups +31%, retention -12%", having dropped everything else. The number is load-bearing enough to be kept.

What does erode is scope, and normative force. Scope restrictions go roughly twice as fast: 61% survive at the same compression against 83% for the negative outcome. And within the corrupted memory itself, separating the finding from the instruction shows the split directly: at the consolidation stage 60/60 chains state the negative outcome and 37/60 still state a prohibition; after four re-summarisations, 60/60 still state the outcome and only 22/60 state a prohibition. What consolidation removes is not the fact that discounting hurt retention. It is the sentence saying not to do it anyway.

Consequences for the threat model, including ours. The corruption Studies 1–3 use — deleting an entire negative caveat — is not what this process produces. A benign-compression threat model should be built on the operators compression actually applies: loss of scope, and loss of normative constraint while the supporting number survives. Those are more dangerous than they sound, because a lesson that retains its numbers looks well-evidenced. Our own Study 3 says such a record draws less verification, not more.

We report this as a limitation on Studies 1–3 rather than repairing it here. The allocation results stand: they concern which beliefs an agent checks given a corrupted one, and hold for any corruption with the same decision geometry. What does not stand is the claim that this particular corruption arises on its own.

A pre-registered rule that failed. We registered that two scorers — a keyword rule and the model scorer — must agree. That rule is mis-specified and we report it rather than its output. Disagreements occur precisely where one scorer sees a lost qualifier, so conditioning on agreement censors informatively and returns 100% survival at every generation by construction. The keyword rule also turned out to have very low specificity on scope restrictions, firing on any mention of the segment term, so it functions only as an upper bound. We report the model scorer, validated against the unqualified controls, as primary, and state the bound rather than presenting the agreeing subset as an estimate.

8 Discussion: memory systems need a verification scheduler

The results describe a control problem that current memory architectures do not expose. A retrieval layer answers *which records are relevant to what the agent is doing*. Nothing answers *which records are worth spending scarce provenance checks on*. In our environment those two rankings are close to opposites: the belief most costly if wrong is the one outside the current plan, and that is exactly the one the budget does not reach.

Making the second ranking explicit needs metadata the first does not: how many consolidation generations a record has passed through, when its provenance was last actually followed, how broad its scope claim is relative to the evidence behind it, and which future contexts could make it load-bearing. Study 4 gives a concrete signal for the third: a record that retains a number while losing its scope or its prohibition has drifted in the direction that matters, and the drift is detectable by comparing generations, which a consolidation pipeline is in a position to do and an agent reading the final summary is not.

We are wary of the term *verification debt*, but the accounting it suggests is the useful part. A durable belief that survives many consolidations without its provenance ever being followed accumulates risk that no single session sees, because each session rationally spends its budget on its own plan. Study 1 shows that behaviour is not a defect of any one model — it is what every model we tested does, and it is locally correct. The fix is therefore unlikely to be a better-behaved agent. It is a scheduler outside the agent that can spend verification against the store's accumulated exposure rather than against today's decision.

Steering works, partially. One system-prompt sentence naming no memory and no direction moves allocation substantially (Table 1, odds ratio 13.2), and Study 2a shows that this reaches behaviour one decision later through what it caused the agent to recover. But it does not equalise models, and in our replay the model with the weakest allocation still walks into the corrupted direction a quarter of the time under the instruction. Prompt-level triage is worth writing and is not a substitute for scheduling.

9 Limitations

One scenario. Every episode instantiates the same synthetic growth-agent situation with six memories and five actions. We deliberately did not add a second domain: built in the time available it would have been a paraphrase of the first, and generality claimed from two structurally identical scenarios is worse than generality not claimed. Domain replication and larger memory pools are the obvious next step and we have not taken it.

Small memory pool, perfect archive. Six records, explicit ids, and a provenance store that always answers correctly and instantly. Real stores are larger, slower, sometimes missing, and sometimes themselves stale. We expect all of these to sharpen the allocation problem, but we have not shown it.

Study 1 is a re-analysis. Its episodes were collected earlier under their own pre-registration. The confounds we identified in its clean arm (Study 3) and in its two-phase follow-up (Study 2) are why Studies 2–4 exist; where Study 1's reported interpretation and our new evidence conflict, we have said so in the section concerned rather than restating the earlier reading.

Causal scope. Studies 2a and 2b identify the effect of *holding* recovered provenance on the next decision. Neither identifies why models allocate as they do, and nothing here tests whether allocation is stable across tasks or sessions — so we avoid calling it a trait.

Behaviour, not harm. We measure which direction the agent chooses and at what scale. Under a strict pre-registered harm rule requiring an unhedged, unverified, corrupted-evidence rollout, no episode in this work or its predecessor qualifies: models hedge verbally while still choosing the direction. That is itself a warning about self-reported uncertainty as a safety metric, and it means our outcome is exposure, not damage.

10 Conclusion

Provenance links do not make an inherited memory store safe; following them does, and following is budgeted. We find that agents spend that budget on the beliefs behind the plan they already have — with no counterexample in 1020 episodes — and that what the budget happens to recover causally determines what they do when the situation later shifts, by roughly ninety percentage points under two independent randomised designs. We also find that a silently deleted caveat attracts less scrutiny than an irrelevant hedge, and that real consolidation mostly does not delete the negative finding — it strips scope and normative force while keeping the numbers that make the lesson look well-evidenced. Retrieval relevance and verification priority are different rankings. Memory systems currently compute only the first.

References

Pritam Dash, Tongyu Ge, Aditi Jain, et al. From untrusted input to trusted memory: A systematic study of memory poisoning attacks in LLM agents. *arXiv preprint arXiv:2606.04329*, 2026.

Zhengru Fang, Senkang Forest Hu, Zhonghao Chang, et al. Inference-time budget control for LLM search agents. *arXiv preprint arXiv:2605.05701*, 2026.

Joshua Klayman and Young-Won Ha. Confirmation, disconfirmation, and information in hypothesis testing. *Psychological Review*, 94(2):211–228, 1987.

Junchi Liao. Auditing provenance sensitivity in LLM agent action selection. *arXiv preprint arXiv:2607.20827*, 2026.

Yuxiang Lin, Zihan Wang, Mengyang Liu, et al. BAGEN: Are LLM agents budget-aware? *arXiv preprint arXiv:2606.00198*, 2026.

Pranav Singh. When does belief-based agent memory help? reliability-conditional updating and provenance-capped poisoning defense. *arXiv preprint arXiv:2606.22030*, 2026.

Gou Tan, Zhensu Sun, Jieke Shi, et al. AgentChaos: Chaos engineering for agent systems via programmatic fault injection. *arXiv preprint arXiv:2608.06790*, 2026.

Daniel Wang and Andrew Xu. AllocBench: Measuring online tool allocation capability in LLM agents. *arXiv preprint arXiv:2607.23332*, 2026.

Yiqi Wang, Jiaqi Zhang, Taotao Cai, et al. From agent traces to trust: A survey of evidence tracing and execution provenance in LLM agents. *arXiv preprint arXiv:2606.04990*, 2026.

Peter C. Wason. On the failure to eliminate hypotheses in a conceptual task. *Quarterly Journal of Experimental Psychology*, 12(3):129–140, 1960.

Weiwei Xie, Shaoxiong Guo, Fan Zhang, et al. MemEvoBench: Benchmarking safety risks from memory misevolution in LLM agents. *arXiv preprint arXiv:2604.15774*, 2026.

Jinlong Yang. Heteroskedastic signals in budgeted LLM verification: Structural heterogeneity limits optimization gains. *arXiv preprint arXiv:2606.15841*, 2026.

Wenhao Yuan, Chenchen Lin, Jian Chen, et al. Belief-guided inference control for large language model services via verifiable observations. *arXiv preprint arXiv:2604.27536*, 2026a.

Wenhao Yuan, Chenchen Lin, Jian Chen, et al. Verify before you commit: Towards faithful reasoning in LLM agents via self-auditing. *arXiv preprint arXiv:2604.08401*, 2026b.